

The Radial Lamb–Oseen Similarity Class: Uniqueness, Exact Particle Angles, and Dissipation Boundaries

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2 September 2026

Abstract

We classify the bounded-at-origin, finite-circulation, radial forward-self- similar solutions of the planar viscous vorticity equation: the signed Lamb–Oseen Gaussian is the entire declared class. Beyond exact Eulerian and Lagrangian formulas, we separate the zero-age, inviscid, origin-particle, long-time, recurrence, and infinite-energy boundaries without extending uniqueness to arbitrary vortices.

1 Declared class and complete classification

On $\mathbb{R}^2$ consider

$$\partial_t \omega + u \cdot \nabla \omega = \nu \Delta \omega, \quad \nabla \cdot u = 0, \quad \text{curl } u = \omega, \quad (1)$$

with the decaying planar Biot–Savart velocity. Fix $\Gamma \in \mathbb{R}$, $\nu > 0$, and $\tau = t + \tau_0 > 0$, where $\tau_0 \geq 0$.

Theorem 1 (Radial forward-similar uniqueness). *Suppose a classical radial solution of (1) has*

$$\omega(x, t) = \tau^{-1} F(\xi), \quad \xi = |x|/\sqrt{\tau}, \quad (2)$$

where $F \in C^2([0, \infty))$ is bounded at the origin and $\int_0^\infty |F(\xi)| \xi \, d\xi < \infty$. If $\int_{\mathbb{R}^2} \omega \, dx = \Gamma$, then

$$\omega(x, t) = \frac{\Gamma}{4\pi\nu\tau} \exp\left(-\frac{|x|^2}{4\nu\tau}\right). \quad (3)$$

Thus (3) is the entire declared similarity class, including $\Gamma = 0$.

Proof. Radial vorticity has radial gradient, whereas its Biot–Savart velocity is tangential; hence $u \cdot \nabla \omega = 0$. Substitution of (2) into the heat equation yields

$$\nu(F'' + \xi^{-1}F') + F + \frac{\xi}{2}F' = 0. \quad (4)$$

Multiplying by ξ and integrating once gives

$$\nu\xi F'(\xi) + \frac{\xi^2}{2}F(\xi) = C. \quad (5)$$

The origin hypotheses make both left-hand terms tend to zero, so $C = 0$. Therefore $F = Ae^{-\xi^2/(4\nu)}$. The normalization $\Gamma = 2\pi \int_0^\infty F(\xi)\xi \, d\xi = 4\pi\nu A$ fixes A and proves (3). $\square$

The scope of Theorem 1 is essential: it classifies every profile satisfying (2) and the displayed regularity and integrability conditions, not arbitrary vortex filaments or arbitrary solutions of (1). The classical Lamb–Oseen mechanism is source owned by Oseen; we claim no historical priority.

2 Velocity and exact Lagrangian dynamics

Polar circulation reconstructs the velocity without an unrecorded constant:

$$u_r = 0, \quad u_\theta(r, t) = \frac{\Gamma}{2\pi r} \left(1 - e^{-r^2/(4\nu\tau)}\right) \quad (r > 0), \quad u(0, t) = 0. \quad (6)$$

Indeed $2\pi r u_\theta = 2\pi \int_0^r s \omega(s, t) ds$; near zero, $u_\theta = \Gamma r / (8\pi\nu\tau) + O(r^3)$.

Proposition 1 (Every particle orbit). *For $r_0 > 0$, radius is invariant. Put $a = r_0^2/(4\nu)$ and*

$$\mathcal{F}_a(\tau) = \tau - \tau e^{-a/\tau} - a \operatorname{Ei}(-a/\tau). \quad (7)$$

Between positive ages $\tau_s < \tau_t$,

$$\theta(\tau_t) - \theta(\tau_s) = \frac{\Gamma}{2\pi r_0^2} [\mathcal{F}_a(\tau_t) - \mathcal{F}_a(\tau_s)]. \quad (8)$$

Here θ denotes a continuous real-valued lift of polar angle, not only its class modulo 2π . The origin is a fixed particle. If $\Gamma \neq 0$ and $r_0 > 0$, then

$$\theta(t) = \frac{\Gamma}{8\pi\nu} \log \tau + O(1) \quad (\tau \rightarrow \infty). \quad (9)$$

Proof. Equation (6) gives $\dot{r} = 0$ and

$$\dot{\theta} = \frac{\Gamma}{2\pi r_0^2} (1 - e^{-a/\tau}).$$

Direct differentiation shows $\mathcal{F}'_a = 1 - e^{-a/\tau}$, proving (8). The continuous origin value proves the fixed-point claim. Finally $1 - e^{-a/\tau} = a/\tau + O(\tau^{-2})$, which gives (9) because $a/r_0^2 = 1/(4\nu)$. $\square$

3 All moments, norms, and exact dissipation

The change of variable $y = r^2/(4\nu\tau)$ evaluates the whole parameter family, rather than a finite selection.

Proposition 2. *For every integer $k \geq 0$ and every real $1 \leq p < \infty$,*

$$\int_{\mathbb{R}^2} |x|^{2k} \omega(x, t) dx = \Gamma k! (4\nu\tau)^k, \quad (10)$$

$$\|\omega(\cdot, t)\|_p^p = \frac{|\Gamma|^p}{p(4\pi\nu\tau)^{p-1}}. \quad (11)$$

Also $\|\omega\|_\infty = |\Gamma|/(4\pi\nu\tau)$ and

$$\int_{\mathbb{R}^2} \omega^2 dx = \frac{\Gamma^2}{8\pi\nu\tau}, \quad (12)$$

$$\int_{\mathbb{R}^2} |\nabla \omega|^2 dx = \frac{\Gamma^2}{16\pi\nu^2\tau^2}, \quad (13)$$

$$\frac{d}{dt} \int_{\mathbb{R}^2} \omega^2 dx = -2\nu \int_{\mathbb{R}^2} |\nabla \omega|^2 dx. \quad (14)$$

Proof. Polar integration and $y = r^2/(4\nu\tau)$ reduce (10) to $\int_0^\infty y^k e^{-y} dy = k!$ and (11) to $\int_0^\infty e^{-py} dy = 1/p$. The supremum occurs at the origin. The case $p = 2$ gives (12); differentiating (3) radially and integrating gives (13). Differentiation in t closes (14). $\square$

4 Boundary atlas and recurrence obstruction

The formulas remain useful only if singular endpoints are not silently substituted into a positive-parameter theorem.

Boundary	Status	Exact conclusion
$\Gamma = 0$	included	$\omega = u = 0$ at every positive age.
$\tau_0 > 0$	included	Smooth finite- <i>enstrophy</i> initial profile.
$\tau_0 = 0$	weak trace	$\omega(\cdot, t) \rightharpoonup \Gamma \delta_0$ as $t \downarrow 0$; smooth for $t > 0$.
$\nu \downarrow 0$	weak limit	Vorticity tends to $\Gamma \delta_0$; velocity tends off the origin to the point vortex.
$r_0 = 0$	separate	The continuous velocity fixes the origin; formula (8) is only for $r_0 > 0$.
$\tau \rightarrow \infty$	asymptotic	Positive-radius angular drift is logarithmic as in (9).
$p > 1$	obstruction	For $\Gamma \neq 0$, every finite L^p norm decreases strictly with age.
$R \rightarrow \infty$	warning	Whole-plane kinetic energy diverges logarithmically.

Positive viscosity is the theorem domain; $\nu = 0$ is never inserted into (3). The L^p law implies that no nonzero vorticity state in this family can recur at two distinct ages. This fluid-state statement does not mislabel the nonautonomous angular motion as a primitive periodic orbit.

The far field $u_\theta = \Gamma/(2\pi r) + o(r^{-1})$ yields the sharp disk asymptotic

$$\frac{1}{2} \int_{|x| < R} |u(x, t)|^2 dx = \frac{\Gamma^2}{4\pi} \log R + O(1). \quad (15)$$

Thus positive-age *enstrophy* and *palinstrophy* are finite, whereas nonzero whole-plane kinetic energy is not. These are different integrability tests.

Source lineage

Oseen’s 1912 line-vortex work is the classical owner token for the profile. Galloway and Wayne provide modern mathematical context for the two-dimensional Oseen vortex and its stability. These citations establish lineage, not a novelty or priority claim for the classical formula.

References

- [1] C. W. Oseen, “Über Wirbelbewegung in einer reibenden Flüssigkeit,” *Arkiv för Matematik, Astronomi och Fysik* 7 (1912), no. 14.
- [2] T. Galloway and C. E. Wayne, “Global stability of vortex solutions of the two-dimensional Navier–Stokes equation,” *Communications in Mathematical Physics* 255 (2005), 97–129. DOI: [10.1007/s00220-004-1254-9](https://doi.org/10.1007/s00220-004-1254-9).